

Quick Check

Lesson 2: How Do Plants Get Rid of Wastes? Nature Walk, Exudation and Guttation Evidence

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.3: Excretion in Plants | Lesson 2

On an early humid morning, a student says, "The droplets along the edges of the arrowroot leaf are just dew." Which observation would best suggest that the droplets come from the plant itself?

- A** The droplets are clear and have no colour
- B** The droplets are seen only in the morning
- C** The droplets are in the same places along the leaf margins and tips, not spread randomly over the leaf
- D** The droplets are wet when touched

A teacher makes a tiny cut on a Euphorbia stem. Thick white milky liquid oozes out and dries to cover the cut. Which waste-removal route does this show?

- A** Exudation of latex through a cut
- B** Diffusion of gases through lenticels
- C** Transpiration through stomata
- D** Guttation through hydathodes

During the nature walk, learners collect fallen yellow leaves and see leaf scars on a branch. Why can the falling of old leaves count as a way a plant gets rid of wastes?

- A** Leaves fall only because of the season, so wastes are not involved
- B** Wastes stored in the old leaves are removed from the plant when the leaves drop
- C** Falling leaves reduce transpiration by closing the stomata
- D** Falling leaves release gum and latex through the leaf scar

In the guttation experiment, one well-watered seedling is covered with a bell jar. A second seedling is left uncovered in a drier spot. What is the purpose of the uncovered seedling?

- A** To show that seedlings do not need water
- B** To compare and show whether high humidity makes guttation droplets appear
- C** To measure how much water the covered seedling absorbs
- D** To make sure the covered seedling gets enough light

Guttation droplets form at leaf edges on a cool, humid night after the soil has been watered well. Which explanation is correct?

- A** The leaves sweat to cool down, as humans do
- B** Roots absorb water faster than transpiration can remove it in humid air, so liquid water is forced out at hydathodes
- C** The stomata open wide and release a lot of water vapour
- D** Dew from the air falls and collects on the leaf edges

Which statement correctly contrasts guttation and transpiration?

- A** Guttation loses liquid water with dissolved substances through hydathodes, mainly at night or early morning. Transpiration loses water vapour through stomata, mainly in the day.
- B** Guttation releases gum and resin from cuts, while transpiration releases latex.
- C** Guttation loses water vapour through stomata in the day. Transpiration loses liquid water through hydathodes at night.
- D** Guttation and transpiration both lose only water vapour, but guttation happens in the roots.

After the nature walk, groups write new questions on the Driving Question Board. Which question is about wastes that are neither gases nor liquids and cannot easily evaporate or ooze out?

- A** Why does Euphorbia latex dry quickly at a cut?
- B** Who buys gum arabic, and for how much?
- C** What happens to wastes that cannot evaporate or ooze out easily?
- D** Why do guttation droplets appear mostly at night?

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).